

Ryan Davis

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Education

University of Michigan, M.S.E. Computer Engineering

Dec 2026 – May 2028

University of Michigan, B.S.E. Computer Engineering

Aug 2023 – May 2027

- GPA: 3.94/4.00

- Awards: EECS Entrepreneurship Award, ZLI Scholar, Applebaum Scholar, Clara Mara Scholar, MBC Finalist

Skills

System Verilog, Verilog, C++, C, CUDA, Python, Assembly, Git, OpenCL, Rust, MATLAB, JavaScript, tcl

Relevant Coursework

Adv. Comp. Arch., Comp. Arch., GPU Arch., Compilers, Algorithms, Comp. Org., Digital Logic, Digital ICs

Work Experience

University of Michigan | Instructional Aide, EECS470 (Computer Architecture)

August 2026 – Present

- Author exam and homework problems covering out-of-order execution, memory hierarchy, and branch prediction. Run weekly discussion sections and hold office hours.
- Building course infrastructure: AI-assisted debug tooling for RTL and testbench failures, OoO pipeline simulator visualizing ROB/RS/LSQ state cycle-by-cycle, migrating EDA flow off Synopsys to open-source.

Qualcomm | CPU Design Intern

May – August 2026

- Owned RTL design and verification of the C1000 datacenter CPU's end-to-end bidirectional DDR address translation pipeline, translating between system addresses and physical target DRAM coordinates.
- Designed interleaving schemes for lower latency accesses and load-balanced traffic across resources. Created configurable LLCC/MC hashing, rank/bank/row remapping, and topology-aware address reconstruction.

Qualcomm | SoC Design Intern

May – August 2025

- Developed RTL for the AI200 AI accelerator chip, optimizing the CMSR (Central Memory Setup & Repair) core that distributes memory redundancy and configuration data to every core on the SoC.
- Implemented RTL to achieve 4x to 8x lower latency data transfers from CMSR to receiving cores by eliminating redundant subsystem-specific data transfer and interfacing with Network on a Chip (NoC).
- Engineered traffic optimization (packet-trimming) within the CMSR core. Implemented on-the-fly parsing and selective transmission of repair-vectors, reducing latency and dynamic power per operation.

MaizeTix | Co-Founder + President

January 2024 – Present

- Built a ticket marketplace for UM sports tickets (maizetix.com) that processed \$1,500,000+ of ticket transactions for 17,000+ verified UM student users. Architected novel automated ticket order fulfillment system.

Projects

Superscalar Out-of-Order RISC-V CPU

- Built fully synthesizable OoO RISC-V CPU in SystemVerilog with ROB/RS/LSQ, prefetching, tournament branch prediction, early tag broadcast, banked I-cache, non-blocking D-cache, byte-level store-to-load-forwarding.
- Drove RTL timing closure from 13ns to 9ns by analyzing critical paths and applying pipelining, parallel datapath lookups, high-fanout signal staging, and combinational logic restructuring.

L1 Cache Simulator for Pipelined Processor

- Developed cache simulator in C to analyze hit rates across memory access patterns and evaluate the performance impact of varying cache capacities, associativities, block sizes, write-allocate behaviors, and eviction policies.

Activities

Michigan Club Golf Team, Computer Architecture Club, Chess Club, Michigan Math Meet